Independent Replication of *Quantum Simulated Annealing*

(arXiv:0712.1008 — Somma, Boixo, Barnum, 2007)

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Abstract

We independently replicate the reproducible core of Somma, Boixo, and Barnum's 2007 *Quantum Simulated Annealing* paper (arXiv:0712.1008): the Szegedy quantum walk $W(M) = R_2 R_1$ constructed from a reversible classical Metropolis Markov chain $M(\beta)$ possesses eigenphases $\pm 2\varphi_j$ with $\varphi_j = \arccos \lambda_j$, giving a phase gap $\Delta_Q \approx 2\sqrt{2\Delta_C}$ in the small-gap limit — a quadratic quantum speedup over the classical mixing time controlled by $\Delta_C = 1 - |\lambda_2(M)|$. On five random $\pm J$ Ising instances (n=4,5,6 spins), three inverse temperatures ($\beta \in \{0.5, 1.0, 2.0\}$), we verify: (i) detailed balance to machine precision, (ii) the coherent Gibbs state $|\pi^{1/2}\rangle$ is exactly fixed by W (residual $< 10^{-15}$), (iii) $\Delta_Q = 2 \sin \varphi_1$ agrees with W's numerical phase gap to $< 10^{-6}$, and (iv) the constant $c := \Delta_Q / \sqrt{\Delta_C}$ lies in $[2.68, 2.83]$ across all 15 (instance, β) rows, tightly bracketing the theoretical $2\sqrt{2} = 2.828$. **Verdict: REPLICATED.** Author list corrected from task brief: the paper is by Somma, Boixo, and *Barnum* (not Knill/Ortiz).

1 Paper metadata (verified)

- **Title:** Quantum Simulated Annealing.
- **Authors:** R. D. Somma (Perimeter Inst.), S. Boixo (LANL / UNM), H. Barnum (LANL). *Corrected from the QC-200 task brief, which listed "Somma, Boixo, Knill, Ortiz". The PDF's title page shows Somma, Boixo, Barnum only.* Somma & Ortiz did coauthor a related earlier paper (arXiv:0706.1146); the 0712.1008 paper cited here is the Somma–Boixo–Barnum one.
- **arXiv:** 0712.1008v1 (submitted 2007-12-06, quant-ph).
- **DOI (journal):** Somma, Boixo, Barnum, Knill, PRL 101, 130504 (2008) (an expanded version added Knill; this arXiv v1 is the 3-author preprint).
- **PDF SHA-256:** see `report/artifacts_summary.md`.

2 Paper summary

Classical simulated annealing (SA) implemented via discrete-time MCMC requires $O(1/\delta)$ Markov-chain queries, where δ is the minimum spectral gap of the Metropolis stochastic matrices $M(\beta_k)$ used along an annealing schedule. Somma, Boixo, and Barnum construct a quantum algorithm — Quantum Simulated Annealing (QSA) — that instead requires $O(1/\sqrt{\delta})$ (with additional polylog(d/ϵ^2) factors), achieving a *quadratic* speedup in the gap parameter. The construction has three pillars:

1. Represent each reversible Metropolis chain $M(\beta)$ as a *Szegedy bipartite quantum walk* $W(M) = R_2R_1$ on an edge Hilbert space $\mathcal{H}\otimes\mathcal{H}$. The two reflections R_1 (about span$\{|\sigma,0\rangle\}$) and R_2 (about span$\{U_X^\dagger U_Y|0,\sigma\rangle\}$) are built from the isometries X, Y that encode column-square-roots of M.

2. Spectral relation (Eqs. 11–14 of the paper): if λ_j are the eigenvalues of $H = X^\dagger Y$ (equal to those of M under detailed balance) and $\varphi_j = \arccos\lambda_j$, then the eigenphases of $W(M)$ are $\pm 2\varphi_j$. Consequently the *quantum phase gap* $\Delta_Q = \min_{j\neq 0}|1 - e^{2i\varphi_j}| \approx 2\sqrt{2\Delta_C}$ in the small-gap limit, since $\Delta_C = 1-\lambda_1 \approx \varphi_1^2/2$.

3. Phase estimation + Zeno projection along a slow annealing schedule $\beta_k = k\Delta\beta$ projects the quantum state at each step onto the coherent Gibbs state $|\phi_0(\beta_k)\rangle = \sum_\sigma \sqrt{\pi_{\beta_k}(\sigma)}|\sigma\rangle$; the runtime of each PEA is set by the walk's phase gap, giving the overall $O(\log^3(d/\epsilon^2)/(\sqrt{\delta}\epsilon^2))$ scaling.

3 Claims table

ID	Claim	Type	Testable in small numpy sim?	Tested here?
C1	Detailed balance: $M[y,x]\pi(x) = M[x,y]\pi(y)$ for Metropolis with acceptance $\min(1, e^{-\beta\Delta E})$.	prerequisite lemma	yes	yes ($< 10^{-18}$)
C2	Eigenvalues of $H = X^\dagger Y$ equal eigenvalues of M (reversible case).	spectral identity	yes	yes (implicit via Δ_Q-vs-λ_2 match)
C3	Eigenphases of $W(M) = R_2R_1$ are $\pm 2\varphi_j$, $\varphi_j = \arccos\lambda_j$.	main structural result	yes	**yes** (Δ_Q matches $2\sin\varphi_1$ to $< 10^{-6}$)
C4	Quadratic speedup: $\Delta_Q \gtrsim c\sqrt{\Delta_C}$, $c \approx 2\sqrt{2}$ in small-gap limit.	main quantitative claim	yes	**yes** (c $\in [2.68, 2.83]$ across 15 rows)
C5	Coherent Gibbs state $|\pi^{1/2}\rangle$ is (via X) fixed by $W(M)$.	structural	yes	**yes** ($\|W\psi - \psi\| < 10^{-15}$)
C6	Full QSA runtime $O(\log^3(d/\epsilon^2)/(\sqrt{\delta}\epsilon^2))$.	composite algorithmic	partially (needs PEA + Zeno on top)	**no** (out of scope for reproducible-core; see open Q3)
C7	Annealing schedule $\Delta\beta = O(\delta/E_M)$, $\beta_f = O(\log(d/\epsilon^2)/\gamma)$ suffices for ϵ error.	schedule/complexity	partially	**no** (see open Q4)

4 Method (numbered, reproducible)

1. **Environment.** macOS 25.3.0 x64 on CherryRd; Python `/usr/bin/python3` (system) with numpy 2.4.3. No GPU, no external services — pure CPU linalg.

2. **Fetch paper.** `curl -sSL https://arxiv.org/pdf/0712.1008 -o work/paper.pdf` then `pdftotext` for skim (poppler 24.x).

3. **Instance family.** Five random Ising instances, generated with numpy default_rng seeds $\{101, 202, 303, 404, 505\}$ at $n \in \{4, 4, 5, 5, 6\}$ spins. Couplings $J_{ij} \in \{-1, +1\}$ uniform, all pairs.
4. **Metropolis chain.** Single-spin-flip Metropolis on $\{-1, +1\}^n$ with proposal prob $1/n$ per spin, accept $\min(1, e^{-\beta \Delta E})$; explicit $2^n \times 2^n$ column-stochastic M matrix.
5. **Classical gap.** $\Delta_C = 1 - |\lambda_2(M)|$ from `numpy.linalg.eigvals`.
6. **Szegedy walk.** Build isometries $X|x\rangle = \sum_y \sqrt{M_{yx}}|x, y\rangle$ and $Y|x\rangle = \sum_y \sqrt{M_{yx}}|y, x\rangle$; reflections $R_A = 2XX^\dagger - I$, $R_B = 2YY^\dagger - I$; walk $W = R_B R_A$ on $\mathbb{C}^{d^2}$ with $d = 2^n$. Verified $X^\dagger X = Y^\dagger Y = I_d$.
7. **Quantum gap.** Diagonalize W; extract phases via `numpy.angle`; $\Delta_Q = \min_{|\theta|>10^{-9}} |1 - e^{i\theta}|$.
8. **Prediction match.** Compute $\varphi_1 = \arccos |\lambda_2(M)|$; compare Δ_Q to $2 \sin \varphi_1$.
9. **Coherent Gibbs check.** $|\psi_G\rangle = X\sqrt{\pi}$; check $\| W|\psi_G\rangle - |\psi_G\rangle \|$.
10. **Repeat** across $\beta \in \{0.5, 1.0, 2.0\}$ and all five instances $\Rightarrow$ 15 rows of (`delta_c`, `delta_q`, `c_ratio`, `gibbs_fixed_err`).

Full source: `report/evidence/qsa_szegedy.py` (405 LOC), run log: `report/evidence/run.log`, structured output: `report/evidence/qsa_results.json`.

5 Results vs paper

Inst. (n, seed)	β	Δ_C	Δ_Q (this work)	$2\sin\varphi_1$ (predicted)	$c := \Delta_Q/\sqrt{\Delta_C}$	$\|W\psi_G - \psi$
(4, 101)	0.5	2.494e−2	4.439e−1	4.439e−1	2.811	3.4e−
(4, 101)	1.0	7.122e−4	7.547e−2	7.547e−2	2.828	7.9e−
(4, 101)	2.0	3.200e−7	1.600e−3	1.600e−3	2.828	1.3e−
(4, 202)	0.5	2.034e−1	1.209e+0	1.209e+0	2.681	1.5e−
(4, 202)	1.0	8.454e−2	8.048e−1	8.048e−1	2.768	4.4e−
(4, 202)	2.0	1.211e−2	3.103e−1	3.103e−1	2.820	3.7e−
(5, 303)	0.5	1.069e−1	8.999e−1	8.999e−1	2.752	2.6e−
(5, 303)	1.0	8.012e−2	7.844e−1	7.844e−1	2.771	3.5e−
(5, 303)	2.0	7.646e−2	7.670e−1	7.670e−1	2.774	6.3e−
(5, 404)	0.5	1.629e−2	3.595e−1	3.595e−1	2.817	4.5e−
(5, 404)	1.0	3.812e−4	5.522e−2	5.522e−2	2.828	4.2e−
(5, 404)	2.0	1.349e−7	1.039e−3	1.039e−3	2.828	1.2e−
(6, 505)	0.5	4.415e−2	5.877e−1	5.877e−1	2.797	6.5e−
(6, 505)	1.0	7.392e−3	2.427e−1	2.427e−1	2.823	2.9e−
(6, 505)	2.0	1.756e−4	3.748e−2	3.748e−2	2.828	7.3e−
Paper's asymptotic prediction	—	—	$\Delta_Q \to 2\sqrt{2}\sqrt{\Delta_C}$	$\to 2\sqrt{2} = 2.828$	(should be	
This work, min/max over 15 rows	—	—	—	[2.681, 2.828]	$< 1.3 \times 10$	

Every row matches the paper's structural prediction to numerical precision. In particular:

- Column "Δ_Q (this work)" and column "$2\sin\varphi_1$ (predicted)" are equal to ≥ 6 digits in *every* row — this is the exact form of the paper's Eq. (12)/(13) main structural result (C3), reproduced numerically.

- The ratio $c = \Delta_Q/\sqrt{\Delta_C}$ converges to $2\sqrt{2}$ from below as $\Delta_C \to 0$ (rows with $\Delta_C < 10^{-3}$ hit $c = 2.828$ to 3 digits), consistent with the small-gap limit of $c = 2\sin\varphi_1/\sqrt{1-\cos\varphi_1}$ which tends to $2\sqrt{2}$.

- The coherent Gibbs state is fixed by W to floating-point noise across all 15 rows, confirming C5.

6 Per-claim: what worked / what did not

C1 detailed balance — WORKED. Max residual $|M_{yx}\pi_x - M_{xy}\pi_y|$ was $\leq 4.3 \times 10^{-19}$ across all 15 (instance, β) combinations.

C2 spectrum(H) = spectrum(M) — WORKED (implicitly). We do not diagonalize H separately; instead the identity $\Delta_Q = 2\sin(\arccos|\lambda_2(M)|)$ holds to 10^{-6}, which would fail if the singular values of the discriminant D (which are what actually enter the Szegedy phases) did not equal $\{\lambda_j\}$. So C2 is confirmed downstream.

C3 eigenphases $\pm 2\varphi_j$ — WORKED. Predicted vs measured quantum phase gap match to $\leq 10^{-6}$ in every row. The full spectrum of W contained $2\cdot(d-1) = 30, 30, 62, 62, 126$ nontrivial phases (plus multiplicities at 0 and π) for $n = 4, 5, 6$, as the paper predicts.

C4 quadratic speedup with $c \approx 2\sqrt{2}$ — WORKED. Minimum c across 15 rows was 2.681 (well above the task's $c > 0.1$ threshold), maximum was 2.828. Deviation from $2\sqrt{2}$ is fully explained by finite-φ_1 curvature (no anomaly).

C5 coherent Gibbs is a +1 eigenstate — WORKED. Residual $\|W|\psi_G\rangle - |\psi_G\rangle\| < 1.3 \times 10^{-15}$ in every row.

C6 full $O(\log^3/\sqrt{\delta}\epsilon^2)$ QSA runtime — NOT TESTED. Out of scope for reproducible-core; would require implementing phase estimation on W plus the Zeno projection schedule. Deferred to open question Q3.

C7 concrete schedule choice — NOT TESTED. Deferred to open question Q4.

7 Verdict

REPLICATED. All 5 structural claims that make sense at $d = 2^n \leq 64$ / $d^2 \leq 4096$ pass. Specifically, satisfying every condition in the task's REPLICATED criterion:

- $\Delta_Q \geq c\sqrt{\Delta_C}$ with $c > 0.1$: yes, with observed $c \in [2.681, 2.828]$ (27× over threshold).

- Verified across ≥ 3 β values: yes, $\beta \in \{0.5, 1.0, 2.0\}$.

- Coherent Gibbs stationary state verified: yes, residual $< 1.3 \times 10^{-15}$.

The two claims not tested (C6, C7) concern the composite algorithm on top of $W(M)$, not the walk-spectrum backbone that is the reproducible core.

8 Notes on tools & provenance

- Marker/Nougat binaries not present on CherryRd; no central-corpus parse of 0712.1008 exists (checked BVBRC/OSTI/LUCID trees). The `extraction/marker.md` and `extraction/nougat.mmd` artifacts are poppler-`pdftotext` fallbacks explicitly marked as such at the top of each file. The numerical replication does not depend on them — it works from the paper's Eqs. (1)–(16) as read directly from the PDF.
- No LLM inference was invoked for this replication (task is a numerical linear-algebra reproduction). Argo endpoint remains available for judge-panel scoring if the wave later requests it.
- Full source, results, run log, and this LaTeX report are shipped under `report/`; downloaded PDF and pdftotext dumps under `work/`.

Open Questions

Full machine-readable form: `report/open_questions.json`.

Q1. How does the constant c in $\Delta_Q = c\sqrt{\Delta_C}$ behave when detailed balance is broken deliberately (asymmetric Metropolis–Hastings proposals, Glauber heat-bath), and does the small-gap limit $c \to 2\sqrt{2}$ survive perturbatively or collapse?

Q2. At what n does the finite-size c ratio depart materially from the small-gap asymptote $2\sqrt{2}$, and is that departure well-predicted by higher-order corrections in φ_1 alone or does $|\lambda_3|, |\lambda_4|$ ordering create larger corrections at moderate temperature?

Q3. Does the phase-estimation + Zeno projection stack of the paper preserve the $O(1/\sqrt{\Delta_C})$ advantage at realistic PEA precisions, or does the $O(\log^3(d/\epsilon^2)/(\sqrt{\delta}\epsilon^2))$ constant swamp the quadratic gain for hard random Ising instances around $n \sim 20$?

Q4. How do the paper's $\Delta\beta = O(\delta/E_M)$ and $\beta_f = O(\log(d/\epsilon^2)/\gamma)$ bounds materialize for the $\pm J$ random Ising ensemble, and is γ (the gap of the objective E) dominated by the ground-state cluster or by intermediate states?

Q5. Is the specific choice of R_2 (reflecting through $\text{span}\{U_X^\dagger U_Y|0,\sigma\rangle\}$) essential for the coherent Gibbs state to be an exact $+1$ eigenvector, or would a naive Szegedy walk $R_B R_A$ with $R_B = 2YY^\dagger - I$ work equally well for reversible chains — and where do the two constructions differ operationally in a quantum circuit implementation?